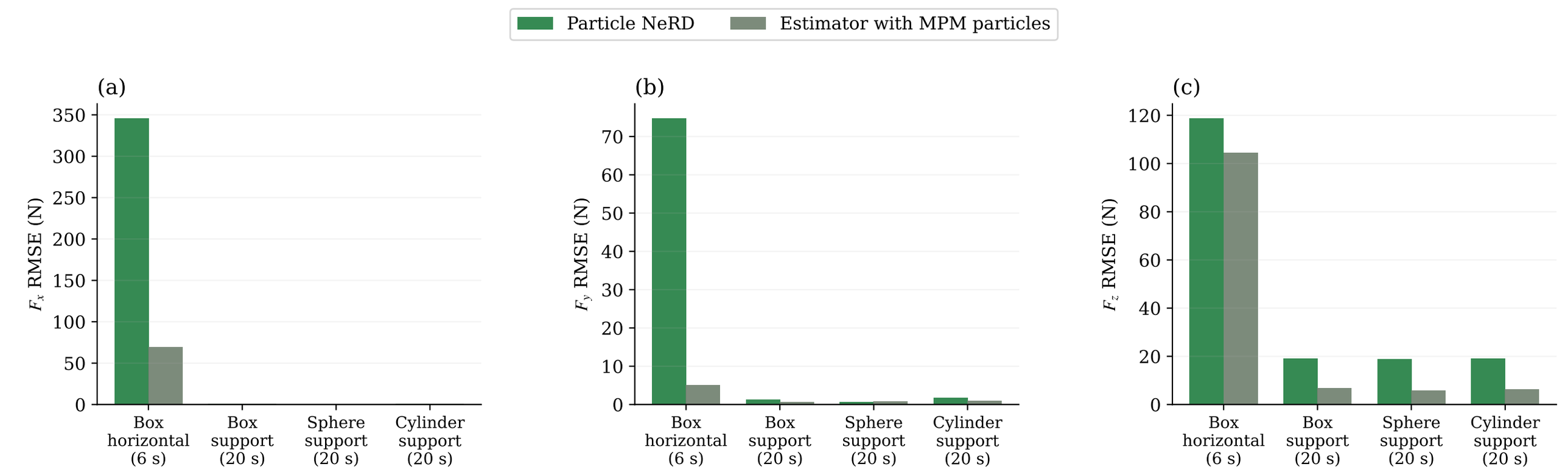

Figure 6: Frozen object-model checkpoint 9,000. (a–c) Fx, Fy, Fz RMSE; (d–f) signed mean errors. Native MPM material force is the reference, excluding rigid-tray force. Signed error is estimate minus MPM. Each case is one raw 200 Hz recording. The gray series substitutes MPM particles into the same estimator; it is not a second network. No force reference exists for the 60-test dataset.